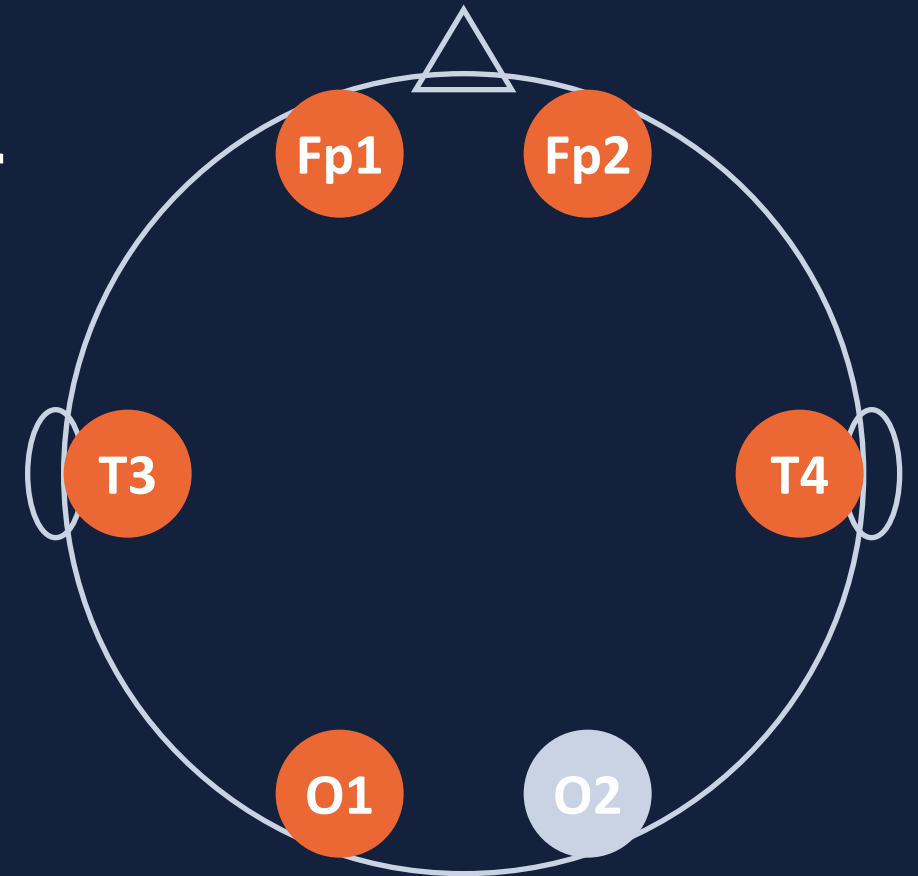


Neurophysiological diagnosis of PTSD

from few-channel dry EEG



The data give the answer away, and that is a trap

0.985

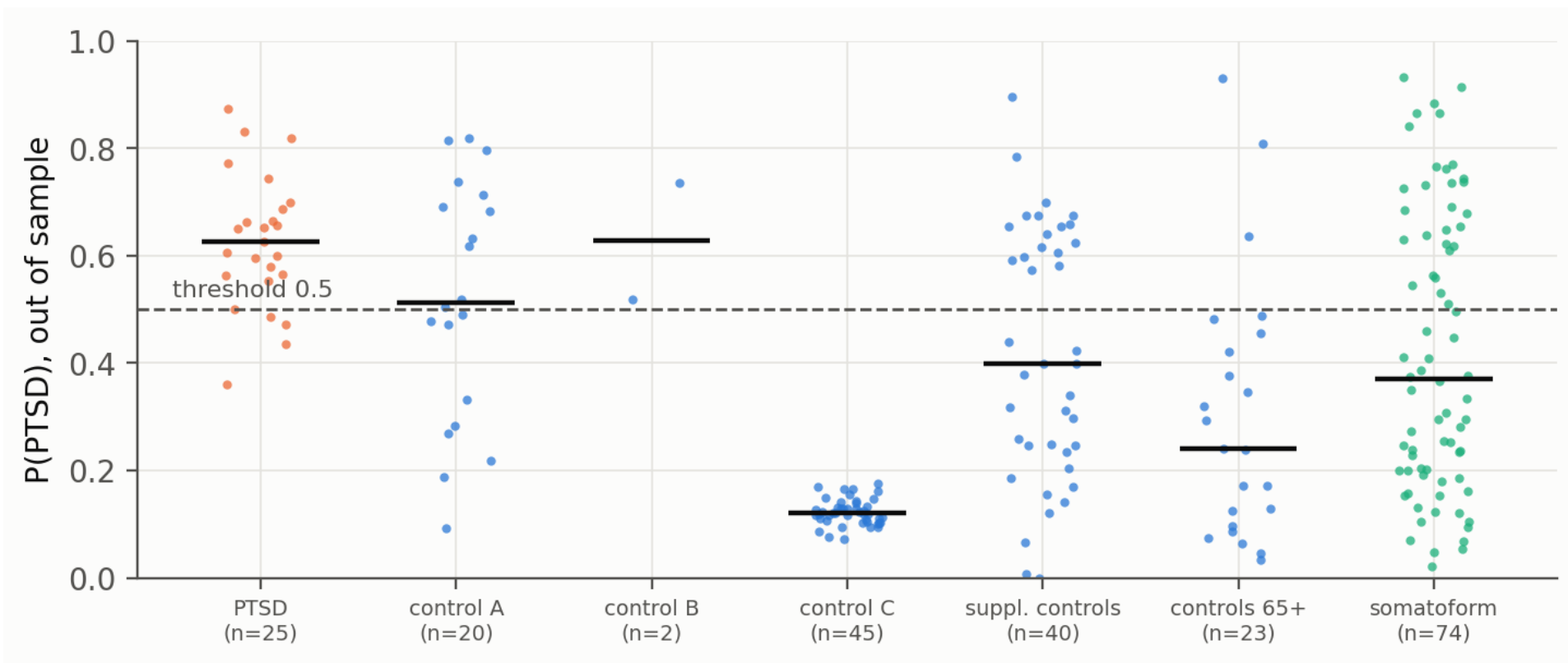
AUC of the rule 'format B means PTSD'
without looking at the signal

- all 25 PTSD patients were exported in format B
- 45 of 67 controls are format C, with no channel correlation and no alpha
- 229 subjects form 221 groups once copied recordings are merged

How we handle it

- 23 resting features: band power, alpha peak, spectral slope; independent of channel gain
- L2 logistic regression with Platt calibration
- leave-one-group-out, every choice made inside the fold
- selection criterion mirrors the official score
- 8 protocols fixed in advance, failed ones included

Out-of-sample result



0.86

AUC, PTSD vs controls < 65
95% CI 0.80–0.92

0.84

sensitivity
at threshold 0.5

0.87

specificity,
controls 65+

0.61

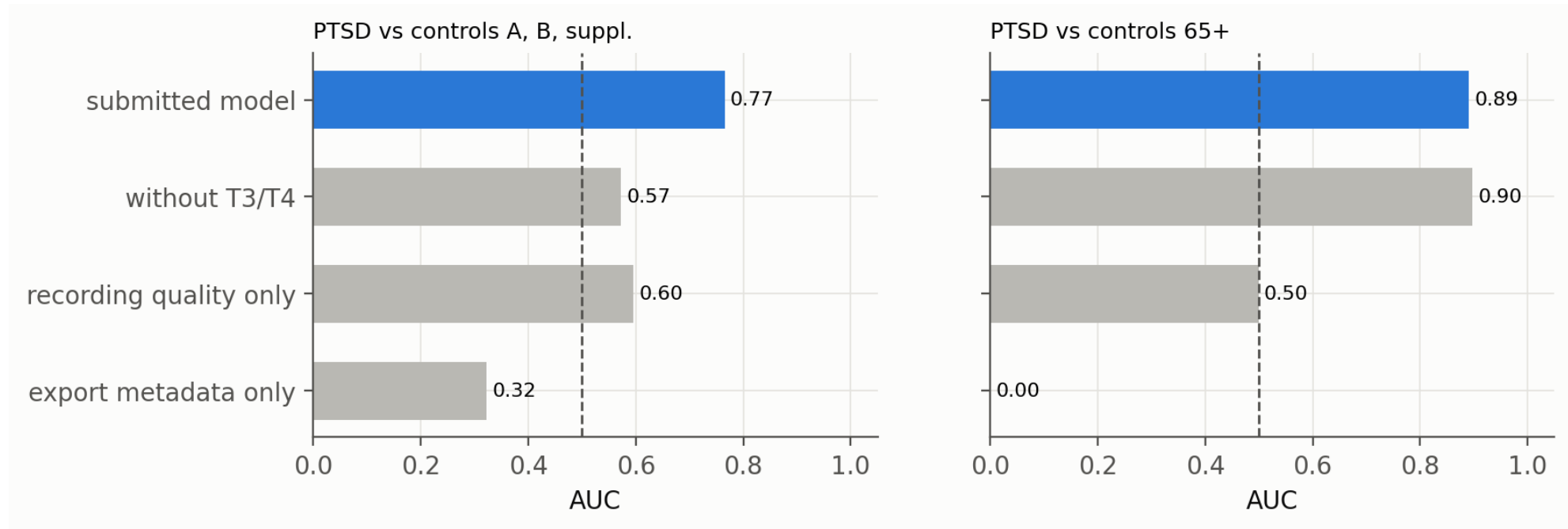
specificity,
somatoform

Features: direction and mechanism

Feature	Lead · band	In PTSD*	Mechanism and caveats
spectral slope	T3, T4 · 3–30 Hz	↓ flatter (0.90 / 0.96)	flatter in PTSD (Kovacevic, 2025); batch-dependent at T3
beta share	T3, T4 · 13–30 Hz	T3 ≈, T4 ↑	T4 shared with somatoform; possible benzodiazepine effect
alpha share	O1 · 8–13 Hz	≈	separates older adults with age-related alpha decline
theta share	Fp1, T3, T4, O1 · 4–8 Hz	↑ slightly (−0.40 / −0.47)	reduced vigilance, poor sleep
spectral slope	O1 · 3–30 Hz	↓ flatter (0.72 / 0.78)	excitation and inhibition balance; same with ageing
alpha peak frequency	O1 · 8–13 Hz	≈ (9.95 / 9.69 Hz)	slows with age
frontal asymmetry	Fp2 – Fp1 · 8–13 Hz	≈	Davidson model; classically F3/F4, no added value

* median in PTSD / control A; arrows relative to young controls; rows ordered by model weight. Only the slope and beta at T4 shift reliably, and both shift the same way in somatoform patients.

What does not explain the result



Format

within format B AUC 0.84;
permutation $p = 0.016$



Age

within a batch $R^2 \approx 0$; P does
not follow age



Behaviour

P does not follow solve
time; with Schulte tables
AUC 0.75



Unseen batch

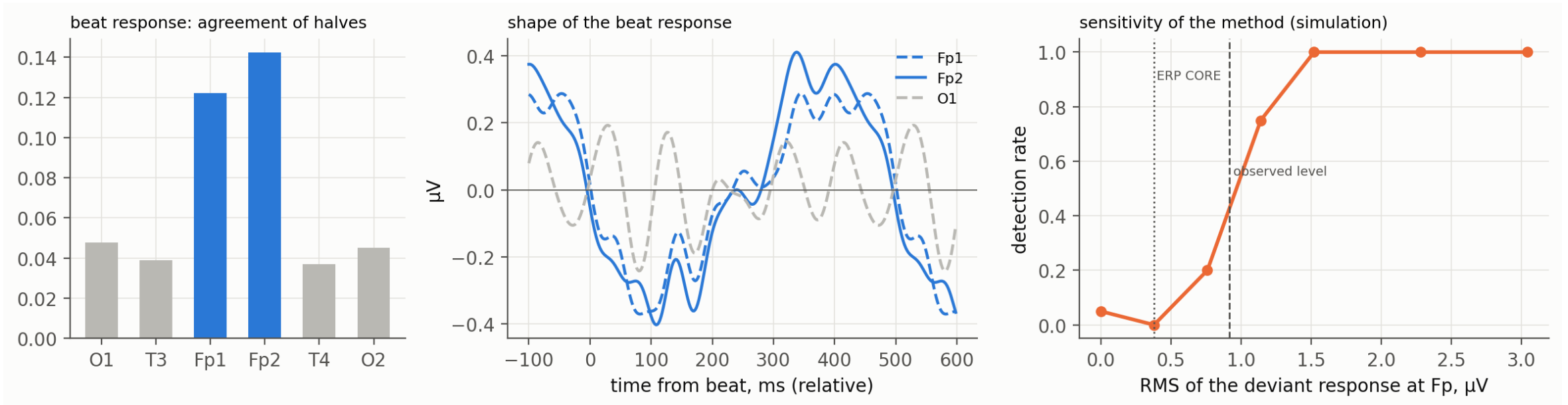
group left out of training:
AUC about 0.1 lower



Limitation

without T3/T4, AUC against
young controls A/B is 0.57

Bonus: auditory response without stimulus markers



shown

response to metronome beats at rest:
 $p = 0.0005$, frontal

validated

method checked on open ERP CORE data
and in a hybrid simulation

not shown

deviant response (MMN/P3a): $p = 0.048$
against 0.025

Conclusions

1

What works

- AUC 0.86 out of sample, sensitivity 0.84
- reliable separation from normal ageing
- no leakage; survives permutation within a format

2

Limitations

- young controls A/B: relies on T3/T4
- somatoform: 0.61, sex not controlled
- cohorts differ beyond the diagnosis

3

Next steps

- combat-exposed controls without PTSD
- one headset and export for everyone
- severity scales, medication, head injury, sleep

Code, notebooks, data audit and clinical interpretation: github.com/igor688-hub/neurodiagnosis-eeg